

Comparison of clinical outcomes and second-look arthroscopic findings after ACL reconstruction using a hamstring autograft or a tibialis allograft

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Received: 29 May 2015 / Accepted: 15 December 2015

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Abstract

Purpose The purpose of this prospective randomized clinical study was to compare the clinical and radiological outcomes, including tibial tunnel widening and the progression of osteoarthritis after ACL reconstruction using a hamstring autograft or a tibialis allograft. In addition, we compared the graft tear and synovial coverage of grafts in patients that underwent the second-look arthroscopy.

Methods Among 184 patients with an ACL injury who underwent ACL reconstruction, 68 patients of autograft group and 64 patients of tibialis allograft group were included for this study after minimum of 2-year follow-up. The Lachman and pivot-shift tests, Tegner activity score, Lysholm knee score, and IKDC score were compared between the two groups. The quadriceps and hamstring isokinetic strengths using dynamometer were also compared. Degree of OA was determined using the Kellgren–Lawrence grading system on the weight-bearing radiographs. In total, 51 patients (26 patients in autograft group and 25 in the tibialis allograft group) underwent the second-look arthroscopy, in which we compared the apparent tear of graft and synovial coverage of grafts.

Results At the final follow-up, there were no statistical significances in the two groups in Lachman and pivot-shift tests (n.s.). The Tegner activity, Lysholm knee score, and IKDC scores were similar in the two groups. Moreover, no significant differences were observed in the muscle power (n.s.). Some patients showed the progression of OA (five in

autograft and four in allograft groups) without intergroup difference (n.s.). Regarding the findings of second-look arthroscopy, although there was no significant difference in graft tear, synovial coverage was better in autograft group than in allograft group.

Conclusion Even though hamstring autografts and tibialis allografts provided good functional outcomes without significant differences, the second-look arthroscopy revealed that hamstring autografts produced better synovial coverage than tibialis allograft.

Level of evidence I.

Keywords Hamstring autograft · Tibialis allograft · Anterior cruciate ligament · Reconstruction · Second-look · Arthroscopy

Introduction

In recent years, there is an increase in the incidence of traumatic anterior cruciate ligament (ACL) disruptions, which is probably due to greater involvement in sports activity [8, 37]. Conservative management used to be preferred over surgery for the treatment of ACL injuries, but it is now known that if not treated appropriately, these injuries may lead to chronic knee instability, meniscal tears, and degeneration of the articular cartilage, and subsequently early-onset osteoarthritis [10, 11, 24]. As a result, for high-demand young patient, surgery is now the preferred form of management for ACL injuries [37].

For successful ACL reconstruction, appropriate patients selection, proper tunnel position, the appropriate graft, stable graft fixation, and adequate rehabilitation are important. However, graft choice remains the subject of debate [7, 9, 15, 25, 32]. Although the patellar tendon

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autograft has been considered the most suitable graft for ACL reconstruction [28], the use of hamstring autografts for ACL reconstruction has increased recently due to improvements made in graft fixation devices. Hamstring tendon autografts for ACL reconstruction have shown better results and lesser donor site morbidity than patellar tendon autografts [14, 27, 29, 36]. However, others have reported that a few hamstring tendon ACL reconstructions showed some deficits of knee flexor power and saphenous nerve damage [1, 35]. Although the allograft has not been frequently used as autograft for primary ACL reconstruction, in a recent study of over 16,000 ACL reconstructions, allografts were used in 42.4 % of primary ACL reconstructions [5, 22]. The primary advantage of allograft tissue is the lack of donor site morbidity. Also additional advantages include a shorter operative time and improved cosmesis [26]. The tibialis anterior and tibialis posterior allografts are being used more frequently than Achilles tendon allografts owing to poor quality of bone block in the latter [3].

Several studies that compared the outcome of autografts and allografts have failed to reach a conclusion as to which is the better, possibly because of different fixation materials and non-prospective study designs [4, 9, 12, 13, 28, 31, 33]. Moreover, many of these studies compared the results of patellar autografts and allograft tendons for ACL reconstruction [12, 17, 28, 33]. However, only a few studies have compared the outcomes of ACL reconstruction using hamstring tendon autografts and tibialis anterior allografts [18, 20, 39], and even fewer have compared the progression of osteoarthritis or the results of second-look arthroscopy after these modalities [20, 38, 39].

The purpose of this prospective randomized study was to compare the clinical and radiological outcomes, including the progression of osteoarthritis and tunnel widening after ACL reconstruction using a hamstring autograft or a tibialis allograft. In addition, we compared the statuses of these two graft types in patients that underwent second-look arthroscopy.

It was hypothesized that ACL reconstruction using a hamstring autograft produces better results in terms of clinical and radiological outcomes and second-look arthroscopic findings than ACL reconstruction using a tibialis anterior allograft.

Materials and methods

Patients with ACL injury, with or without meniscal injury, were included in this prospective study. Patients with fractures on the same side, multiple ligament injuries, and those who underwent revision ACL reconstruction were excluded. Also, patients with radiological evidence of

degenerative changes in either knee with Kellgren–Lawrence grade of worse than two were excluded.

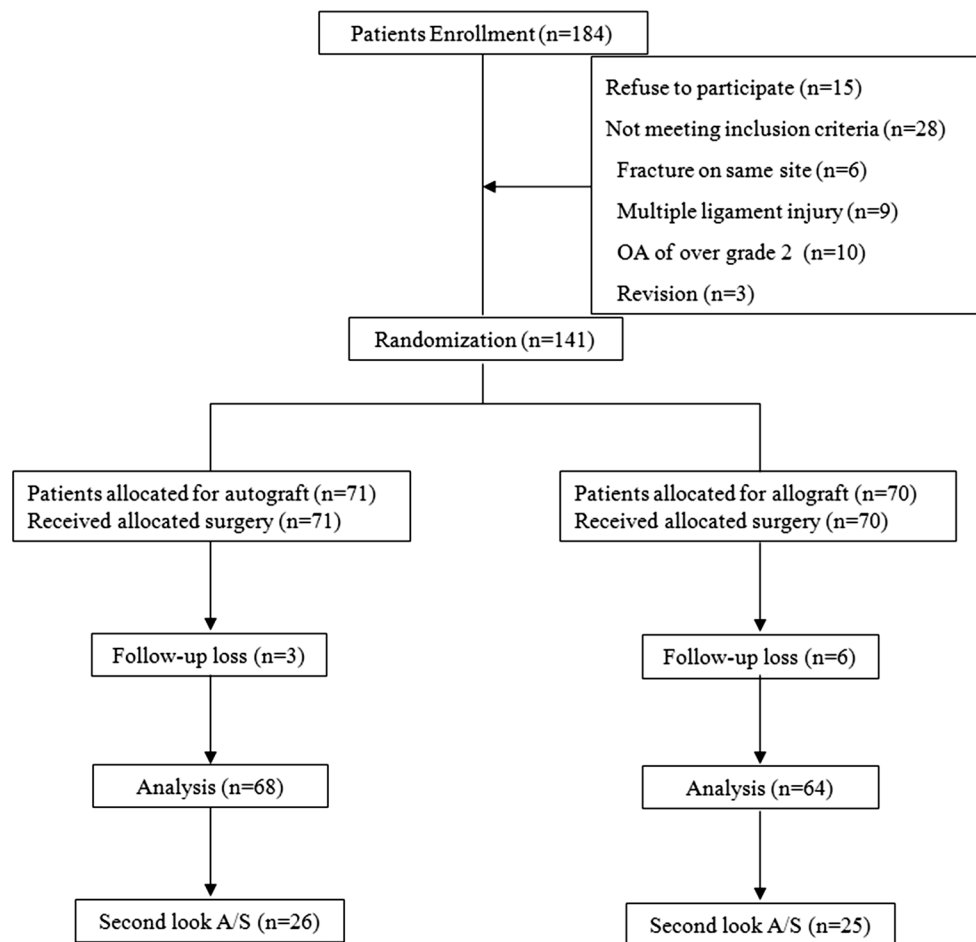
One hundred and eighty-four patients with an ACL injury underwent ACL reconstruction from 2008 to 2011. However, 28 patients did not meet inclusion criteria and 15 patients refused to participate in the study (nine patients refused to randomly use graft), which left 141 patients for this study. Patients were randomized to a hamstring autograft ($n = 71$) or tibialis allograft ($n = 70$) group using the sealed envelope method the day before surgery. However, the patients were blinded for their group information. The written consent to participate in this study was taken from all patients. A flow diagram of CONSORT chart is provided in Fig. 1.

After routine examination of knee using arthroscopy, meniscal repair or meniscectomy was performed when concomitant meniscal injuries were observed. For ACL reconstruction, semitendinosus and gracilis tendons were harvested through longitudinal skin incision over per anserinus and prepared as double-looped autografts (autograft diameter, 6–8 mm). A aseptically harvested, fresh-frozen tibialis allografts (Tissue Banks International, San Rafael, CA, USA) were prepared as single-looped grafts (allograft diameter, 7–8 mm). The proximal end of the graft was fixed with an EndoButton, and the distal ends were prepared with whipstitch sutures. After tibial tunnel preparation using a drill guide within the centre of the ACL insertion, a femoral tunnel was created at the centre of the footprint through the accessory anteromedial portal. An EndoButton CL (Smith & Nephew, Andover, MA, USA) was used for femoral side fixation, in both groups. The graft on the tibial side was fixed at 10°–20° of flexion, using bioabsorbable interference screws (Linvatec, Largo, FL, USA) under 40°N of tension. Additional fixation on the tibial side was performed using a screw and washer.

The rehabilitation protocol used was identical in two study groups. After operation, a knee brace was applied with the knee joint extended at 0°. Quadriceps isometric exercises were initiated immediately after surgery. From 1 day after surgery, 0°–90° flexion exercise was encouraged with gradual increase to full range of motion at 4 and 6 weeks under knee brace. Patients were allowed weight bearing with crutches under knee brace as tolerated. In cases of combined meniscus repair, full weight bearing was delayed for about 6 weeks. In all patients, jogging was usually allowed at 3 months after surgery and sports-specific activities from 6 months.

To evaluate knee laxity, the Lachman and pivot-shift tests were used before surgery and at final follow-up. Lysholm score regarding limp, locking, pain, stair climbing, use of supports, instability, swelling, and squatting was measured to compare the knee function. Tegner activity score and IKDC scores were also used to evaluate knee

Fig. 1 Flow diagram for the Consolidated Standards of Reporting Trials (CONSORT). OA osteoarthritis, A/S arthroscopy



function, and quadriceps and hamstring isokinetic strength was assessed at the velocities of 180°/s with a dynamometer (Biodex system 2; Biodex Medical System, Shirley, New York, USA) at the final follow-up visit after the ACL reconstruction. A standardized application of equipment, data collection, and procedure for warm-up were applied. Before data collection was started, the subjects had performed three practice repetitions at both angular velocities. The uninjured extremity was tested first, and then, the same procedure was performed for the injured extremity. Quadriceps and hamstring isokinetic peak torques of injured extremities were expressed as percentages versus uninjured extremities.

Knee laxity was also evaluated using an instrumented laxity test and a Telos[®] stress device (Austin & Associates, Fallston, MA, USA) at 30° of knee flexion with 20 lb of anterior load applied to the proximal tibia [30]. The difference of anterior translation between affected and contralateral uninjured knee determined the laxity. The degree of OA was determined using the Kellgren–Lawrence grading system. The weight-bearing anteroposterior and lateral radiographs were taken at final follow-up, and numbers of patients with OA progression by more than one grade

versus preoperative grade were noted. The largest diameter of the sclerotic margins of the tibial tunnel was measured on knee anteroposterior and lateral radiographs. And all measurements were corrected for magnification. All values were compared with the known intraoperative sizes of the reamers for tibial tunnel [8]. Radiographic assessments were performed by two orthopaedic surgeons who were not part of the operating team and were blind to the surgical group assignment.

Second-look arthroscopy was performed only in patients who requested tibial fixation screw removal after ACL reconstruction. The risks and benefits of second-look arthroscopy were explained to all patients. Fifty-one patients underwent second-look arthroscopy (26 patients in the hamstring autograft group and 25 in the tibialis allograft group). Mean time between ACL reconstruction and second-look arthroscopy was 18.1 months (range 12.5–24.2 months). To evaluate the tear of graft bundles during second-look arthroscopy, we used a modification of the classification system for ACL grafts described by Kondo and Yasuda [16]. According to this system, status of graft tear was graded as A (intact), B (partial tear), or C (total tear) (Fig. 2). In addition, synovial coverage of

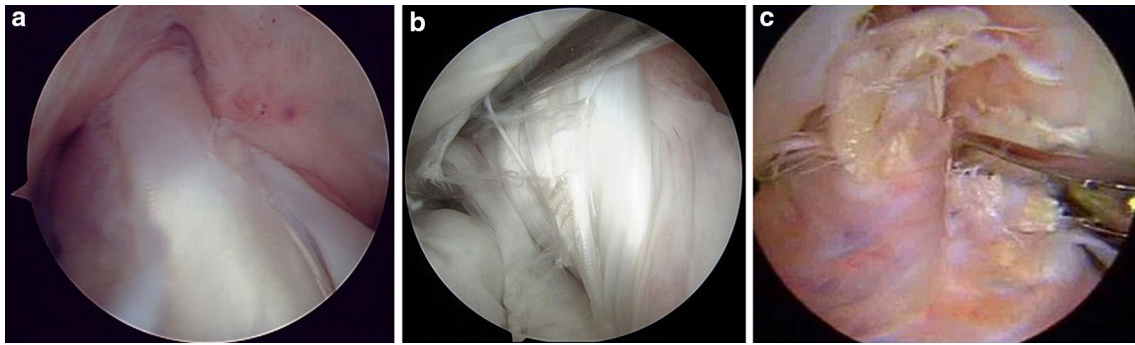
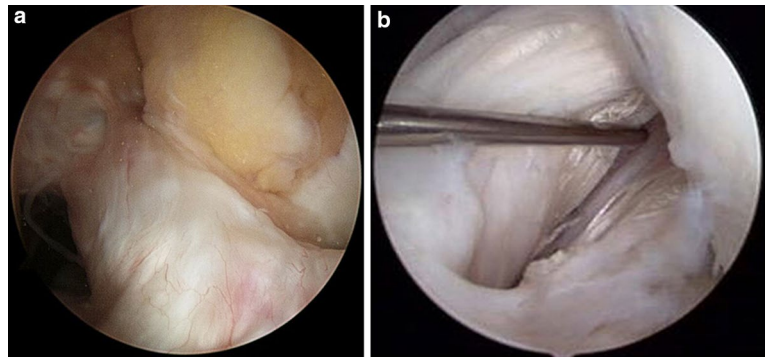


Fig. 2 Types of tear pattern. **a** intact, **b** partial tear, and **c** total tear

Fig. 3 Synovial coverage. **a** good, **b** poor



reconstructed grafts was graded as good (nearly complete or partial coverage; more than 20 %) and poor (no coverage; less than 20 %) (Fig. 3) [2, 20, 39]. Also, chondral lesion was assessed and graded according to the modified outer bridge classification during the second-look arthroscopy and compared the incidence chondral injury between the autograft and allograft groups. This prospective study was approved by Chonnam National University Hwasun Hospital Institutional Review Board (2007–107).

Statistical analysis

The sample size was estimated, and the following primary parameters were chosen to assess power values: a difference in the instrumented laxity test of 1.5 mm (SD, 2.7 mm) and a difference in Lysholm score of four points (SD, 7 points). For these parameters, a sample size of 51 patients per group was sufficient to obtain a power of 0.8. Preoperative and post-operative comparison of Tegner activity scores and instrumented laxity test results was made using the paired sample *t* test, and the inter-group comparisons of these results were made using the independent-samples *t* test. The Pearson χ^2 test was used to determine whether differences were present between autograft and allograft groups with respect to the distribution of combined meniscal injuries, Lachman test results,

pivot-shift test results, IKDC objective examination ratings, and second-look arthroscopic findings. Statistical analysis was performed using SPSS ver. 19.0 for Windows (SPSS, Chicago, IL, USA). Statistical significance was accepted for *p* values <0.05.

Results

Due to follow-up loss (three in hamstring autograft and six in tibialis allograft groups), 68 patients [61 men and 7 women, median age 30 years (range 15–62 years)] of hamstring autograft group and 64 patients [59 men and 5 women, median age 24 years (range 13–52 years)] of tibialis allograft group were included for this study. Demographic characteristics are shown in Table 1.

Comparison of combined meniscal injuries and surgical procedures between two groups is shown in Table 2. The two groups were similar in terms of the incidence of meniscectomy and repair (n.s.).

All patients recovered full range of motion at final follow-up. Comparison of knee laxity and clinical outcome between two groups is summarized in Tables 3 and 4.

Isokinetic dynamometer testing showed that flexor and extensor muscles in affected knee joints in the autograft group had lower peak torques than unaffected knee joints at

Table 1 Comparison of patient demographic data between two groups

Variable	Autograft (<i>n</i> = 68)	Allograft (<i>n</i> = 64)	<i>p</i> value
Age (mean, range)	30 (15–62)	24 (13–52)	n.s.
Sex (male/female)	61/7	59/5	n.s.
Time to surgery (month, range)	1.3 (0.1–108.3)	1.7 (0.1–50.1)	n.s.
Follow-up duration (month, range)	32.8 (28.7–52.1)	34.5 (25.3–59.5)	n.s.

Table 2 Comparisons of combined meniscal injuries and surgical procedures between two groups

Variable	Autograft (<i>n</i> = 68)	Allograft (<i>n</i> = 64)	<i>p</i> value
Meniscal injury	42	39	n.s.
LM	15	11	
MM	18	24	
LM + MM	9	4	
Meniscectomy (partial/subtotal)	28 (20/8)	22 (18/4)	n.s.
Repair	14	17	n.s.

LM lateral meniscus, MM medial meniscus

Table 3 Comparison of knee stability in the two study groups

Grade	Autograft (<i>n</i> = 68)		Allograft (<i>n</i> = 64)		<i>p</i> value
	Preop	Final F/U	Preop	Final F/U	
<i>Lachman test</i>					
0	0	33	0	30	n.s.
1	0	31	0	28	
2	47	3	49	4	
3	21	1	15	2	
<i>Pivot-shift test</i>					
0	0	42	0	31	n.s.
1	0	24	0	30	
2	43	2	44	1	
3	25	0	20	2	

Preop preoperation, Last F/U last follow-up

final follow-up (7.9 % deficit in flexor and 19.2 % deficit in extensor power). Similarly, affected knees had lower peak torques than unaffected knees in the allograft group (9.8 % deficit in flexor and 21.2 % deficit in extensor power), and no significant intergroup difference was evident (n.s.).

Instrumented laxity test, performed using the Telos® device, showed an improvement in the hamstring autograft group from 10.8 to 2.2 mm and in the tibialis allograft group from 10.2 to 2.5 mm. However, the difference between two groups was not significantly different at final follow-up. Radiological outcome about comparison of osteoarthritic change after ACL reconstruction between two groups is shown in Table 5. However, there was no

significant intergroup difference. Among them, three patients of autograft group and three patients of allograft group performed meniscectomy in primary ACL reconstruction. The tibial tunnel widening was increased in anteroposterior (2.1 mm in autograft vs. 2.6 mm in allograft; n.s.) and lateral radiographs (1.9 vs. 2.1 mm, respectively, n.s.).

The results of second-look arthroscopy are summarized in Table 6. No significant difference intergroup was found with respect to tears, but synovial coverage was significantly better results in autograft group than in allograft group ($p = 0.03$). Also, chondral lesion was found in both the autograft and allograft groups (Table 7). While the lesions were mainly located on the patellofemoral or medial compartment, no cartilage lesions occurred in the lateral compartment. There were no statistical significances in the two groups (n.s.).

Concerning complications, ACL reconstruction revision was performed in a single patient in both groups. Lachman, pivot-shift, and radiological stability evaluation tests revealed abnormal laxity of more than grade 2 in one patient in hamstring autograft group and in four patients in the tibialis allograft groups. Two patients with an early post-operative infection required antibiotic treatment in the tibialis allograft group. Sensory nerve injury occurred in four patients in hamstring autograft group and in two patients in tibialis allograft group (Table 8).

Discussion

The most important finding of this study was that even though both hamstring autografts and tibialis allografts used in ACL reconstruction provided good clinical and radiological outcomes, the second-look arthroscopy revealed that hamstring autografts produced better synovial coverage than tibialis allograft. Although comparative studies on the uses of autografts and allografts for ACL reconstruction have shown similar clinical and radiological outcomes [7, 12, 13, 19, 28, 31–34], most were conducted using patellar or Achilles tendons [7, 12, 13, 28, 33]. Furthermore, only few studies [19, 34] have evaluated the outcome of ACL reconstruction, especially regarding second-look arthroscopic findings, using soft tissue hamstring autografts or tibialis allografts. In this study, we compared

Table 4 Comparison of clinical results after ACL reconstruction between two groups

Variable	Autograft (<i>n</i> = 68)	Allograft (<i>n</i> = 64)	<i>p</i> value
LK score	96 (range 67–100)	93 (range 73–100)	n.s.
Tegner activity score	5 (range 2–9)	5 (range 3–8)	n.s.
IKDC			
A	41 (60.3 %)	29 (45.3 %)	n.s.
B	24 (35.3 %)	31 (48.4 %)	
C	3 (4.4 %)	4 (6.3 %)	
D	0 (0 %)	0 (0 %)	

ROM range of motion, LK score Lysholm knee score, Preop preoperation, Last F/U last follow-up, IKDC International Knee Documentation Committee

Table 5 Comparison of osteoarthritic changes after ACL reconstruction between two groups

K–L grade	Autograft (<i>n</i> = 68)		Allograft (<i>n</i> = 64)		<i>p</i> value
	Preop	Last F/U	Preop	Last F/U	
0	58	53	55	51	
1	10	13	9	12	
2	0	2	0	1	
3	0	0	0	0	
4	0	0	0	0	
No. of OA progression	5 (7.35 %)		4 (6.25 %)		n.s.

K–L grade Kellgren–Lawrence grade, Preop preoperation, Last F/U last follow-up, OA osteoarthritis

Table 6 Second-look arthroscopic findings in the study groups

	ACL reconstruction (<i>n</i> = 51)		<i>p</i> value
	Autograft (<i>n</i> = 26)	Allograft (<i>n</i> = 25)	
<i>Status of graft</i>			
Intact	19 (73.1 %)	18 (72.0 %)	n.s.
Partial tear	6 (23.1 %)	6 (24.0 %)	
Total tear	1 (3.8 %)	1 (4.0 %)	
<i>Synovial coverage</i>			
Good	22 (84.6 %)	14 (66.0 %)	0.03
Poor	4 (15.4 %)	11 (44.0 %)	

ACL reconstruction using hamstring autograft and tibialis allograft in terms of clinical and radiological outcomes with a minimum 2-year follow-up. Our data demonstrated that we were unable to identify any differences between the autograft and allograft in terms of clinical or radiological outcomes including OA progression and tibial tunnel

Table 7 Comparison of cartilage status at second-look arthroscopy

	Patellofemoral compartment (n.s.)		Medial compartment (n.s.)	
	Autograft (<i>n</i> = 26)	Allograft (<i>n</i> = 25)	Autograft (<i>n</i> = 26)	Allograft (<i>n</i> = 25)
0	21	20	23	23
1	3	2	2	1
2	2	3	1	1
3	0	0	0	0
4	0	0	0	0

widening. However, findings of second-look arthroscopy did reveal better synovial coverage in the autograft group compared to the allograft group.

Several studies [19, 23, 34] have reported no differences in stability after ACL reconstruction between a hamstring autograft and a tibialis allograft. Similarly, we did not find any difference between the two in terms of the Lachman test, the pivot-shift test, or the instrumented laxity test. Moreover, number of patients showed grade II or III residual laxities by Lachman test (four in autograft and six in allograft groups), which concurs with the findings of other literature conducted on single-bundle ACL reconstructions [5, 34].

The similar results of functional outcomes regarding Tegner activity score and IKDC score have been reported in many studies [19, 23, 34]. In our study, functional outcomes in terms of Tegner activity score, Lysholm knee score, and IKDC score were significantly improved after ACL reconstruction regardless of both groups. However, the Tegner activity or Lysholm knee and IKDC scores showed no statistical significant differences between two groups at final follow-up (n.s.). At the final follow-up, the median Lysholm knee score and Tegner score of patients were improved to 96 (range 67–100) and 5 (range 2–9) in the autograft group and 93 (range 73–100) and 5 (range 3–8) in the allograft group, respectively. These results did not show statistical differences between the two groups. Sun et al. [34] found no significant difference between autograft and allograft in their IKDC scores. They reported that 93 % of autograft and 91 % of allograft groups showed normal or near normal activity at the final follow-up. This study result of IKDC score (normal or near normal: 95 % of autograft and 93 % of allograft group) was comparable to their results.

Lipscomb et al. [21] reported no significant differences between these two graft types in terms of knee flexor strength after ACL reconstruction, but Carter et al. [6] reported significantly lower knee flexor strengths for hamstring tendon autografts than tibialis tendon allografts. In the present study, evaluations of muscular strengths of knee joints at final follow-up showed similar decreases in flexor

Table 8 Major complications of ACL reconstruction in the two groups

	Autograft (<i>n</i> = 68)	Allograft (<i>n</i> = 64)	<i>p</i> value
Revision due to graft rupture	1	1	
+3 Lachman or higher	1	2	
+3 Pivot-shift or higher	0	2	
Infection	0	2	
Sensory nerve injury	3	4	
Total	5 (7.35 %)	11 (17.2 %)	n.s.

and extensor strengths in both groups and no intergroup differences. Only few previous study has investigated the incidences of osteoarthritis after ACL reconstruction performed using hamstring autografts or tibialis allografts. Lawhorn et al. [18] reported 4 % of autograft and 10 % of allograft groups showed mild osteoarthritis after 2-year follow-up in ACL reconstruction. In the present study, no significant difference was found between two groups with respect to the progress of osteoarthritis. The tibial tunnel widening was observed in both groups without statistical significance at the final follow-up.

Yoon et al. [39] reported better results for synovial coverage following ACL reconstruction using a hamstring tendon autograft than a tibialis tendon allograft based on second-look arthroscopic findings. In the present study, rupture rates of grafted tendons were similar in two groups, although the hamstring autograft group showed better synovial coverage than the tibialis allograft group, which suggests healing process of the graft is more rapid in hamstring tendon autografts than in allografts. Moreover, there was no statistical significant difference in chondral injury between two groups at the second-look arthroscopy finding.

The major limitations of this study are that only graft status in selective patients was evaluated because some patients refused to perform the second-look arthroscopy, and that the follow-up duration was relatively short. These limitations hampered objective assessments of the maturity of grafted tendons during ligament development. Moreover, the evaluation system of graft might be subjective although several studies have used for the evaluation of graft in ACL reconstruction. Although the allograft was prepared as 7–8 mm diameter to reduce bias that might be made due to graft thickness, difference of diameter between autograft and allograft can also lead to potential bias. And the autograft group was a little bit older than the allograft group without statistical significance. In addition, the exclusive use of plain radiographs and the short follow-up prevented accurate evaluation of progress of

osteoarthritis. Nevertheless, this study is significant in the sense that it is prospective randomized designed study to compare the clinical and radiological outcomes after ACL reconstruction using a hamstring autograft or a tibialis allograft with the same fixation device and the statuses of these two graft types in patients that underwent second-look arthroscopy.

Conclusion

Both hamstring autografts and tibialis allografts used in ACL reconstruction provided good clinical and radiological outcomes without statistical significant difference between two study groups. However, second-look arthroscopy revealed that hamstring autograft showed a significant better synovial coverage than tibialis allograft.

Acknowledgments This study was supported by a grant of the National Research Foundation of Korea funded by the Ministry of Education, Science and Technology (2014R1A1A2059147).

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